

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

AI

Section Id :	640653141350
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	7
Number of Questions to be attempted :	7
Section Marks :	25
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653398276
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532166097 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR

PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406537043202. ✓ Yes

6406537043203. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653398277

Question Shuffling Allowed :

No

Question Number : 2 Question Id : 6406532166098 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question

ASK FOR PRINTED
GRAPH SHEETS
2 PAGES TWO-SIDE PRINT

Options :

6406537043204. ✓ Printed graph sheets were provided on time.

6406537043205. ✗ Printed graph sheets were provided late.

6406537043206. ✗ Printed graph sheets were not provided.

6406537043207. ✓ I used the graph sheets.

6406537043208. ✗ I did not use graph sheets.

Sub-Section Number :

3

Sub-Section Id :

640653398278

Question Shuffling Allowed :

No

Question Id : 6406532166099 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (3 to 8)

Question Label : Comprehension

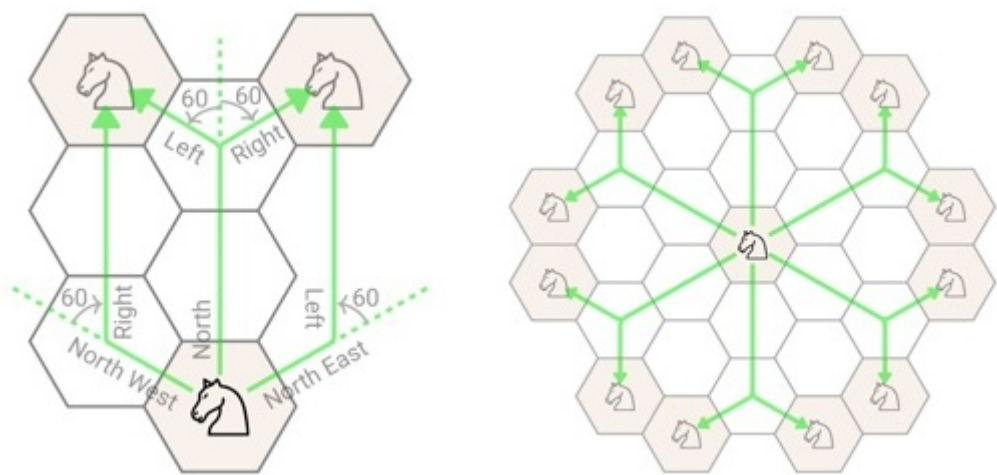
STATE SPACE SEARCH

A gameboard is made of hexagonal tiles, allowing 6 directions of movement from each tile: north, south, north-east, north-west, south-east, and south-west.

A knight can move in two equivalent ways: take **two** steps in a chosen direction, turn 60 degrees left or right, and take **one** final step; or equivalently take **one** step in a chosen direction, turn 60

degrees left or right, and take **two** final steps.

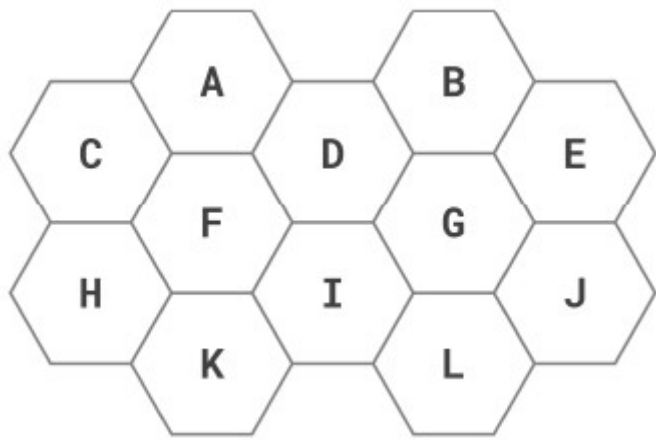
This rule generates 12 move choices as illustrated below.



A move is valid if its starting and landing tiles exist, even if the intervening tiles are missing.

Problem Statement:

Figure shows a gameboard with 12 positions (tiles A to L). MoveGen takes a position and returns a list of valid move choices in alphabetical order, e.g., $\text{MoveGen}(A) = [E, L]$.



Each position is a regular hexagon with unit sides, where opposite vertices are 2 units apart and opposite sides are $\sqrt{3}$ units apart.

Take the distance between two positions as the Euclidean Distance between the center points of corresponding hexagons. For example, $d(A,A) = 0$, $d(A,B) = 3$, $d(A,F) = \sqrt{3}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 3 Question Id : 6406532166100 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Design a state representation for the gameboard, enter its description in the textbox as succinct as possible.

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

Gameboard position

Board position

Question Number : 4 Question Id : 6406532166101 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

The number of unique states in the gameboard state space is _____ .

Enter an integer.

NO SPACES, TABS, DOTS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Number : 5 Question Id : 6406532166102 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

MoveGen(G) is _____ .

Enter a comma separated list of positions.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: A,B,C,D

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

C,H

Question Number : 6 Question Id : 6406532166103 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

d(C,G) is _____ .

Enter a decimal number rounded to one decimal place.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: 42.1

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4.6

Question Number : 7 Question Id : 6406532166104 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

What is true about the gameboard state space?

(Note: a path is a sequence of one or more moves.)

Options :

6406537043213. ✓ At least one state has no path from another state.

6406537043214. ✗ Every state has a path to every other state.

6406537043215. ✓ Every state has at most two neighbours.

6406537043216. ✗ Every state has at least 12 neighbours.

Question Number : 8 Question Id : 6406532166105 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Can a knight complete a TSP tour on the gameboard, visiting all the positions?

Options :

6406537043217. ✖ Yes

6406537043218. ✔ No

6406537043219. ✖ Cannot be determined

Sub-Section Number : 4

Sub-Section Id : 640653398279

Question Shuffling Allowed : No

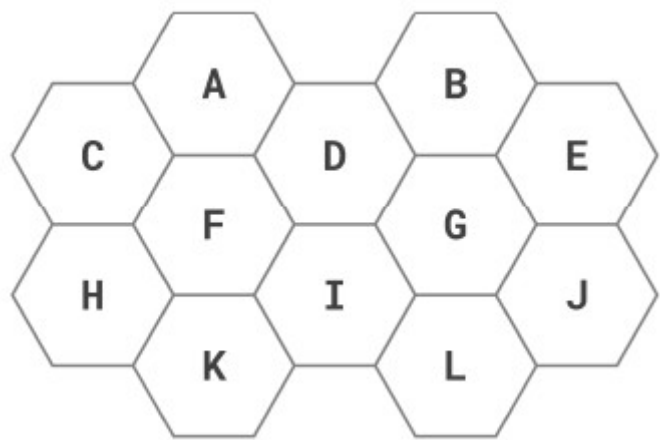
Question Id : 6406532166106 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (9 to 16)

Question Label : Comprehension

SEARCH

Figure shows a gameboard with 12 positions (tiles A to L). MoveGen takes a position and returns a list of valid knight move choices in alphabetical order, e.g., MoveGen(A) = [E,L]. Valid knight move choices are as described in the STATE SPACE SEARCH section.



Each position is a regular hexagon with unit sides, where opposite vertices are 2 units apart and opposite sides are $\sqrt{3}$ units apart.

Take the distance between two positions as the Euclidean Distance between the center points of corresponding hexagons. For example, $d(A,A) = 0$, $d(A,B) = 3$, $d(A,F) = \sqrt{3}$.

There is a knight in position F and no other pieces on the board.

Take F as the start position and C as the goal position.

Use the Euclidean Distance as the heuristic function.

Use alphabetical order to break ties.

Note: When we say a node (a state) is inspected/expanded/refined it means: the node (the state) is picked up from OPEN, and goal test is called, if goal test fails then MoveGen is called and, depending on the algorithm, the neighbours are selectively placed in OPEN.

Note: RemoveSeen procedure will drop neighbours already present in OPEN or CLOSED list.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 9 Question Id : 6406532166107 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

List the first 4 positions inspected by Depth-First Search. List the positions in the order they are inspected. If the algorithm terminates early then list the positions inspected up until termination.

Enter a comma separated list of positions.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

F,E,A,L

Question Number : 10 Question Id : 6406532166108 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Depth-First Search?

Enter the path as a comma separated list of positions.

Enter NIL if a path to the goal is not found.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

F,E,A,L,H,G,C

Question Number : 11 Question Id : 6406532166109 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

List the first 4 positions inspected by Breadth-First Search. List the positions in the order they are inspected. If the algorithm terminates early then list the positions inspected up until termination.

Enter a comma separated list of positions.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

F,E,J,A

Question Number : 12 Question Id : 6406532166110 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Breadth-First Search?

Enter the path as a comma separated list of positions.

Enter NIL if a path to the goal is not found.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

F,J,K,B,C

Question Number : 13 Question Id : 6406532166111 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 1

Question Label : Short Answer Question

List the first 4 positions inspected by Best-First Search. List the positions in the order they are inspected. If the algorithm terminates early then list the positions inspected up until termination.

Enter a comma separated list of positions.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

F,E,A,L

Question Number : 14 Question Id : 6406532166112 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Best-First Search?

Enter the path as a comma separated list of positions.

Enter NIL if a path to the goal is not found.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

F,E,A,L,H,G,C

Question Number : 15 Question Id : 6406532166113 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the path found by Hill Climbing?

Enter the path as a comma separated list of positions.
Enter NIL if a path to the goal is not found.
NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer Format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

NIL

Question Number : 16 Question Id : 6406532166114 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

The heuristic function for the gameboard _____ .

Options :

6406537043227. ✖ defines a maximization problem

6406537043228. ✔ defines a minimization problem

6406537043229. ✖ is monotonic

6406537043230. ✔ is non-monotonic

Sub-Section Number : 5

Sub-Section Id : 640653398280

Question Shuffling Allowed : No

Question Id : 6406532166115 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (17 to 21)

Question Label : Comprehension

ALGORITHMS

Answer the given subquestions.

Sub questions

Question Number : 17 Question Id : 6406532166116 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the formulas that are in Conjunctive Normal Form.

Options :

6406537043231. ✖ $(A \wedge B) \vee (\neg B \wedge C) \vee (\neg D)$

6406537043232. ✔ (A

$\vee B) \wedge (\neg B \vee C) \wedge (\neg D)$

6406537043233. ✓ $A \wedge B \wedge \neg C \wedge D$

6406537043234. ✓ $A \vee B \vee \neg C \vee D$

6406537043235. ✗ $A \vee (B \wedge (\neg C \vee \neg D))$

Question Number : 18 Question Id : 6406532166117 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Perturbation method(s) used for TSP tour creation is/are _____ .

Options :

6406537043236. ✓ 2-city exchange operator

6406537043237. ✓ 3-edge exchange operator

6406537043238. ✗ single-point crossover operator

6406537043239. ✗ Nearest Neighbour Heuristic

Question Number : 19 Question Id : 6406532166118 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If a genetic algorithm prematurely converges to a homogeneous suboptimal population, then which modification will prevent such convergence?

Options :

6406537043240. ✗ Increasing the selection pressure.

6406537043241. ✗ Decreasing the mutation rate.

6406537043242. ✓ Increasing the mutation rate.

6406537043243. ✗ Truncating low rank parents before crossover.

Question Number : 20 Question Id : 6406532166119 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If you initially implement Breadth-First Search using First-In-First-Out Queue and later replace the queue with Max-Priority Queue based on node depth (larger depth equals higher priority), then the memory footprint of the queue would _____ .

Options :

6406537043244. ✗ drop to $O(1)$

6406537043245. ✗ practically remain the same

6406537043246. ✓ significantly decrease

6406537043247. ✗ significantly increase

Question Number : 21 Question Id : 6406532166120 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

In the Simulated Annealing schedule, if the temperature parameter T drops to absolute zero very early in the search process, the algorithm will move from the current node N to a random neighbour X _____ .

Options :

6406537043248. ✖ only if N is better than X

6406537043249. ✔ only if X is better than N

6406537043250. ✖ unconditionally (resulting in a random walk)

Sub-Section Number : 6

Sub-Section Id : 640653398281

Question Shuffling Allowed : No

Question Id : 6406532166121 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (22 to 24)

Question Label : Comprehension

TSP

Figure shows a distance matrix for 5 cities, use it to construct TSP tours.

	A	B	C	D	E
A	-	50	36	28	30
B	50	-	82	36	58
C	36	82	-	50	32
D	28	36	50	-	22
E	30	58	32	22	-

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 22 Question Id : 6406532166122 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The given TSP is a _____. (CAUTION: May require bull work.)

Options :

6406537043251. ✔ Euclidean TSP

6406537043252. ✖ non Euclidean TSP

Question Number : 23 Question Id : 6406532166123 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Construct a tour using Nearest Neighbour Heuristic, start from city A. Enter the path representation of the tour in the order it is constructed.

Enter a comma separated list of city names.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer format: A,X,Y

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,D,E,C,B

Question Number : 24 Question Id : 6406532166124 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Use city A as the base (fulcrum) city to complete the savings list then construct the Savings tour. Enter the path representation of the tour starting from city A.

BC	BD	BE	CD	CE	DE
?	?	22	14	34	36

Enter a comma separated list of city names.

NO SPACES, TABS, BRACKETS, PARENTHESIS OR UNWANTED CHARACTERS.

Answer format: A,X,Y

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

A,B,D,E,C

A,C,E,D,B

Question Id : 6406532166125 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (25 to 27)

Question Label : Comprehension

TSP ALGORITHMS

Answer the given subquestions.

Sub questions

Question Number : 25 Question Id : 6406532166126 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

A major drawback of Nearest Neighbour Heuristic for TSP is _____ .

Options :

6406537043255. ✖ it takes exponential computation time

6406537043256. ✖ it cannot handle sparse graphs

6406537043257. ✔ it often leaves a very long edge for the final return leg

6406537043258. ✖ it often returns premature subtours

Question Number : 26 Question Id : 6406532166127 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

What is the core principle used in the Savings Heuristic for TSP?

Options :

6406537043259. ✖ Add the cheapest available edge to the tour.

6406537043260. ✔ Maximize the distance saved in merging two subtours.

6406537043261. ✖ Minimize the number of cities in each subtour.

6406537043262. ✖ Sort the cities by their distance from the base/fulcrum city.

Question Number : 27 Question Id : 6406532166128 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The savings $S(a,b)$ for two cities a and b relative to base/fulcrum city n is _____ , where $C(x,y)$ is the distance between cities x and y .

Options :

6406537043263. ✖ $C(a,b) - (C(n,a) + C(n,b))$

6406537043264. ✖ $(C(n,a) + C(n,b)) - 2 \cdot C(a,b)$

6406537043265. ✖ $(C(n,a) \cdot C(n,b)) / C(a,b)$

6406537043266. ✔ $(C(n,a) + C(n,b)) - C(a,b)$